

24. Sharpening of Image Using High-Boost Masks

Aim

To perform **image sharpening using a High-Boost Mask** by enhancing the high-frequency components of an image.

Procedure

1. Read the input image.
2. Convert the image to grayscale.
3. Apply a smoothing filter such as Gaussian blur to obtain a blurred image.
4. Subtract the blurred image from the original image to obtain the **mask**.
5. Choose a high-boost factor $A \geq 1$.
6. Add the scaled mask to the original image.
7. Normalize the pixel values to the range **0–255**.
8. Display the original and sharpened images.
9. Save the sharpened output image.

Formula

High-Boost Filtering:

where $A \geq 1$.

- $A = 1 \rightarrow$ Standard Laplacian-type sharpening
- $A > 1 \rightarrow$ High-boost sharpening

Program

```
import cv2

import numpy as np

# Read input image
```

```
img = cv2.imread("input.jpg")

# Check image

if img is None:

    print("Error: Image not found")

    exit()

# Convert to grayscale

gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

# Apply Gaussian blur

blur = cv2.GaussianBlur(gray, (5, 5), 0)

# High-boost factor

A = 2

# High-boost filtering

sharpened = cv2.addWeighted(gray, A, blur, -(A - 1), 0)

# Display images

cv2.imshow("Original Image", gray)

cv2.imshow("High-Boost Sharpened Image", sharpened)

# Save output
```

```
cv2.imwrite("high_boost_output.jpg", sharpened)
```

```
cv2.waitKey(0)
```

```
cv2.destroyAllWindows()
```

Input



Output:



Result

The image was successfully sharpened using the **High-Boost Mask** technique. The edges and fine details of the input image were enhanced, producing a clearer and sharper output image.